

Syed Sulaiman

Electronics & Embedded Hardware Engineer

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OBJECTIVE

Embedded Electronics Engineer with hands-on experience in battery systems, automotive electrical architecture, and safety-critical electronics through Formula Student EV development and robotics product engineering. Experienced in accumulator design, voltage/current sensing, embedded PCB development, HV/LV system integration, and electrical validation. Interested in building reliable and high-performance battery systems for electric mobility applications.

EDUCATION

- **PES University** 2021 - 2025
B.Tech in Electronics and Communications, GPA: 8.77/10.00 Bangalore, India
 - Specialization in Signal Processing and Systems Engineering
 - Minors in Computer Science Engineering
- **Christ Junior College** *Pre-University Education, Grade: 95.6%* Bangalore, India 2021
- **St. Paul's English School** *Secondary Education, Grade: 94.83%* Bangalore, India 2019

EXPERIENCE

- **Systemantics India Pvt Ltd**  January 2025 - Present
Associate Design Engineer - Robotics Bangalore, India
 - Designed and developed embedded electronics subsystems for a collaborative robot, including Raspberry Pi(CM5) based system control, stepper motor controllers, force-torque sensing modules, and LVDT controllers, with end-to-end schematic and PCB design following DFM/DFA principles.
 - Performed circuit design, power budgeting, component selection and tolerance analysis, including buck/boost power supply design with detailed calculations and worst-case analysis; validated through testing and debugging.
 - Designed STM32-based embedded hardware with CAN, SPI, I²C, UART, RS485, and ADC interfaces for sensor acquisition, real-time control, and subsystem communication.
 - Worked on high-speed interfaces including Gigabit Ethernet, USB 3.0, HDMI, and PCIe, considering signal integrity, routing constraints, and PCB stack-up design.
 - Conducted hardware validation and debugging using oscilloscopes and logic analyzers, performing root cause analysis(RCA) to ensure robust performance, reliability, and system integration.
 - Led the development of a jet engine turbine blade metrology and inspection station, integrating LVDT-based dimensional measurement with a machine vision diffuse dome illumination system for automated defect detection.
 - Developed STM32-based embedded firmware for real-time control, communication, and hardware interfacing across multiple subsystems, including closed-loop stepper motor servo control with encoder feedback
- **Artificial Intelligence and Robotics Lab (AIRL), Indian Institute of Science** June 2024 - August 2024
Robotics Intern Bangalore, India
 - Set up and configured a Unmanned Underwater Vehicles (UUVs) ROS-Gazebo simulation environment; conducted a literature review on swarm coordination and developed a multi-agent distributed area-search simulation.
- **Systemantics India Pvt Ltd**  June 2023 - July 2023
Design Intern Bangalore, India
 - Developed STM32 firmware for a stepper motor driver, implementing low-level motor control routines for integration with the robot's gripper mechanism.
 - Performed technical evaluation of 10+ depth camera systems and designed custom PCBs including a Raspberry Pi Camera HAT with ToF sensing and LED flash support for a wrist-mounted collaborative robot vision system.

FORMULA STUDENT EXPERIENCE

- **Electrical Lead, Vega Racing Electric** 2022 - 2025
Formula Student team, PES University 
 - Played a key role in rebuilding the team and developing a fully functional race car from scratch
 - Led the end-to-end development of the vehicle's electrical architecture (high-voltage, low-voltage and VCU), including energy storage, safety systems, and embedded subsystems for an electric race car.
 - Designed and integrated the complete accumulator(Tractive system energy storage), including cell configuration, HV busbar distribution, Contactors Relays(AIRs), voltage and current sensing circuits, fuse and connector selection, isolation strategy, and safety-critical protection systems in compliance with Formula Student EV regulations.
 - Programmed and integrated the Orion BMS with the accumulator, motor controller, and VCU via CAN communication, implementing protection logic, cell balancing during charging and discharging, voltage,current and temperature monitoring, and fault diagnostics.

- Performed DFMEA and root cause analysis across accumulator subsystems, HV /LV interfaces, shutdown circuitry, and system-critical signals to identify electrical failure modes and improve operational safety and reliability.
- Led system-level validation and root cause analysis across HV battery systems, BMS, motor control, LV safety circuits and VCU using oscilloscopes, multimeters, and CAN diagnostics during system bring-up and testing.
- Architected and implemented an ESP32-based vehicle data acquisition and telemetry system integrating 10+ sensors and CAN communication with BMS and Motor Controller, enabling real-time monitoring, onboard logging, and driver visualization via GUI and GPS-enabled ground station.
- Worked closely and established cross-functional integration workflows with mechanical team to translate electrical and safety requirements into manufacturable robust accumulator packaging and HV system integration.

PROJECTS

• Formula Student Electric Vehicle – Safety-Critical Circuits & Low Voltage Architecture

Tools: KiCAD, EasyEDA, LTSpice, MultiSim, Oscilloscopes & Logic Analyzers

- Architected and implemented the complete low-voltage electrical system with load analysis and worst-case design calculations with strong focus on failure mode identification and mitigation to ensure safe, fault-tolerant operation.
- Designed and validated standalone analog safety-critical circuits (no MCU dependency) including:
 - * Shutdown Circuit (SDC) for immediate HV system deactivation under fault(temp/current/voltage) conditions
 - * Brake System Plausibility Device (BSPD) to detect unintended motor torque during braking
 - * Tractive System Active Light (TSAL) voltage sensing circuit for low & high-voltage state indication
 - * Precharge and interlock circuits for controlled and safe high-voltage system activation
 - * Insulation monitoring interface to ensure safe isolation between High and Low-voltage domains
- Developed multiple PCBs using discrete analog design (comparators, op-amps, MOSFETs, relays), incorporating tolerance analysis, component derating, grounding strategy, and EMI/noise mitigation techniques.
- Designed HV–LV interfacing and ensured strict compliance with Formula Student rules, including isolation requirements, System Critical Signal handling, and use of safety-rated components (e.g., UL94-V0 for HV paths).
- Built and validated complete vehicle electrical integration including HV /LV harnessing, interlock connector systems, crimping standards, vibration-resistant electrical joints, ensuring robustness for motorsport environments.

PUBLICATIONS

J=JOURNAL PUBLICATION

- [J.1] **Sulaiman, S. et al. WareOdo: Leveraging warehouse landmarks for correcting odometry in indoor mobile robots.** *Intel Serv Robotics*, Vol. 18, pp. 687–708 (2025). DOI: <https://doi.org/10.1007/s11370-025-00610-4>
- Collaborated with Accio Robotics to develop a landmark-assisted odometry correction framework using EKF sensor fusion and AprilTag global pose correction, achieving 60% drift reduction and state-of-the-art localization accuracy

SKILLS

- **Electronics Software Tools:** KiCad, Autodesk Eagle, EasyEDA, Altium Designer, LTSpice
- **Embedded programming Tools:** STM32CubeIDE, Arduino IDE, FreeRTOS
- **Communication Protocols:** CAN, I2C, SPI, USART, USB, Ethernet, Modbus
- **Modeling & Simulation Tools:** MATLAB, Simulink
- **Programming Languages:** C, C++ , Embedded C, Python, Shell scripting, Java
- **Computer Vision:** OpenCV, PyTorch, TensorFlow, Google Cloud Vision, YOLO
- **Robotics Tools:** ROS, RViz, Gazebo, MoveIt
- **CAD:** Fusion 360

PORTFOLIO

- Detailed engineering projects and Formula Student EV work:- [Portfolio](#) 